

DATE: 09/22/2026
TO: Kid Kidd
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SUBJECT: Project 1

1 INTRODUCTION

As engineers, the documentation of a project is just as crucial as implementing the project itself. The purpose of this project is to Derive engineering and testing requirements. Design, build and test an embedded system that allows you to set the blink intervals of two LEDs independently.

2 PROBLEM STATEMENT

There are 3 parts to this assignment. In Part I, I will derive the High-level Engineering Requirements that correspond to the Customer Requirements given. These are not the high-level engineering requirements I will use in Parts II and III. In Part II, I will derive testing requirements given a different set of high-level requirements along with a set of educational requirements/constraints. In Part III, I will design, build and test the system in Part II.

3 EQUIPMENT

- Arduino Uno R3 development board or clone, e.g. the ELEGOO Uno R3
- LEDs, qty 2
- 200Ω ¼ Watt resistors, qty 2
- 10kΩ ¼ Watt resistor, qty 1
- Arduino IDE 2.3.3 or better
- Laptop or another computer to run the Arduino IDE
- Cable, USB 2.0 female type B to female USB 2.0 type A

4 PART I CUSTOMER REQUIREMENTS -> HIGH-LEVEL ENGINEERING REQUIREMENTS

PROBLEM TO BE SOLVED: From the Customer's requirements in Table 1, derive the high-level technical requirements for the system being designed. These requirements are written in the Customer's language and understandable by the Customer. Typically, these requirements are not specific enough to be used for engineering purposes.

Table 1. Customer Requirements

C1.	The User will be able to choose the blink rate of the LED with a button.
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C2.	The User will choose between 4 rates plus off.
C3.	The blink rates must be set by an operator prior to the system being started.
C4.	Permitted blink rates must be perceivable by the viewer.
C5.	The minimum blink rate must be no slower than once every 5 seconds.
C6	The System must run upon an Arduino Uno R3 compatible board.

5 PART II: DERIVE TESTING REQUIREMENTS

PROBLEM TO BE SOLVED: Given the high-level technical requirements (HL. #) and educational constraints (EC.#) below – and different from the one’s you derived in Part II – derive the testing requirements (labeled T.#) that will allow the company to prove to the customer that The Company has satisfied the contract.

Table 2. High Level technical requirements

HL.1	The System must use at least 2 LEDs
HL.2	The System must use a standard Arduino compatible development board (e.g. the provided ELEGOO Uno R3)
HL.3	The System must communicate with The User via the Arduino IDE serial-monitor port
HL.4	Any use of the serial-monitor in HL.3 must be asynchronous and not affect the blinking of the LEDs.
HL.5	The User must set the blink interval of the LEDs in msec
HL.6	The blink rate of each LED must be constant unless changed by The User

Table 3. Education Constraints

EC.1	The Developer must use the version of gcc used by Arduino IDE 2.3.3 or later for the AVR ATmega328p to cross compile any program used by the System
EC.2	The Developer must have any software used in The System hosted on a GitHub repository. This software does not need to include libraries, macros or other code not developed by The Developer.

EC.2.1	1 The Evaluator must be able to (a) download from the repository any software, documentation and other material needed for their evaluation, (b) cross compile any software, and (c) run that software on The Reference Platform.
EC.3	LEDs must be connected to Arduino digital Pins 2 & 3.
EC.4	The Evaluator must be able to successfully have the system perform the dialog in Table 4
EC.5	The Developer must do bare metal programming, i.e. they cannot use any 3rd party multitasking libraries or programs such as FreeRTOS.

Table 4. Example Dialog (*Red* are the user inputs).

Serial Port I/O	Notes
What LED (1 or 2) <i>2</i>	
What interval (in msec)? <i>600</i>	LED2 starts blinking at an interval of 300ms on and 300ms off. LED1 is unaffected.
What LED (1 or 2) <i>1</i>	
What interval (in msec)? <i>1600</i>	LED1 starts blinking at an interval of 800ms on and 800ms off. LED2 is unaffected.
What LED? (1 or 2)	Waiting for next LED# interval pair

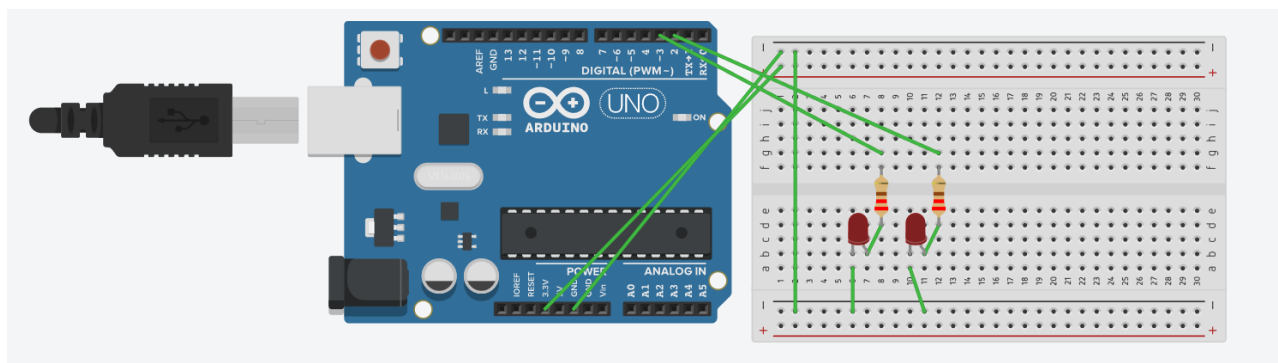
T.1	The Evaluator must visually confirm that two LEDs are present and wired, and that each LED illuminates independently when its corresponding digital pin is driven HIGH.
T.2	The Evaluator must confirm, by inspection of the physical wiring or schematic, that LED1 is connected to digital Pin 2 and LED2 is connected to digital Pin 3.
T.3	The Evaluator must open the Arduino IDE serial-monitor and confirm that prompts and responses matching Table 4's dialog are visible.
T.4	With both LEDs blinking at two different, previously set intervals, the Evaluator must send a new interval to one LED via the serial-monitor, and confirm that the unaffected LED's blink rate does not change or pause during or after the input.
T.4.1	The serial monitor must remain responsive to further input immediately after the change.

T.5	The Evaluator must enter a specific interval (in msec) via the serial monitor for a given LED and measure that LED's actual on/off period using a video timestamp.
T.6	With one LED set to a fixed interval, the Evaluator must send a new interval to the <i>other</i> LED and confirm the first LED's period does not change and continues blinking at its original rate.
T.7	The Evaluator must confirm, via the build output or project configuration file, that the tool chain used to compile the software is the AVR gcc distributed with Arduino IDE 2.3.3 or later.
T.8	The Evaluator must clone the Developer's GitHub repository onto a machine with only Arduino IDE 2.3.3 or later
T.9	The Evaluator must execute the exact dialog sequence shown in Table 4 via the serial monitor and confirm that:
T.9.1	Each prompt text matches the example dialog.
T.9.2	LED2 begins blinking at a 300 ms on 300 ms off rate after the 600ms inputs
T.9.3	LED1 begins blinking at an 800 ms on 800 ms off rate after the 1600ms input, without affecting LED2.
T.9.4	After both inputs, the serial-monitor returns to prompting What LED? (1 or 2)
T.10	The Evaluator must inspect the source code and confirm no 3rd-party multitasking library.

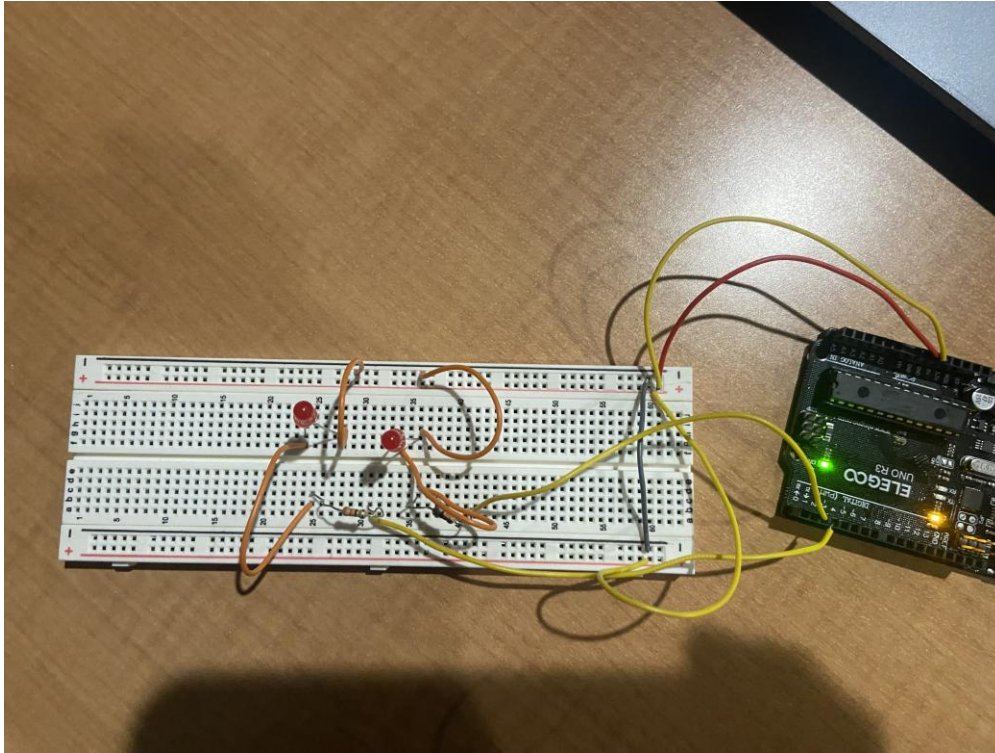
6 PART III: DESIGN, BUILD AND TEST SYSTEM

Schematic:

This schematic was done with Tinker CAD circuits. There were a couple of assumptions that LEDs are standard red LEDs. The board is an Arduino UNO and LED 1 is connected to pin 2 and LED is connected to pin 3.



Hardware Design:



Resistor Values: 330 ohms. I had to use a 330-ohm resistor because I had no 200-ohm resistor.

7 CONCLUSIONS

This project derived high-level requirements from the customer's requirements (Part I), derived testing requirements from a separate set of high-level requirements and constraints (Part II), and used those to design, build, and test a working System (Part III). The completed System independently blinks two LEDs on digital Pins 2 and 3, with intervals set via the serial-monitor without one LED's timing affecting the other. Testing (T.1–T.10) confirmed the System correctly reproduces the example dialog and satisfies HL.1–HL.6, with the only deviation being the use of 330 Ω resistors in place of 200 Ω due to parts availability, which dimmed the LEDs slightly but did not affect functionality. Overall, the System satisfies all specified requirements and constraints.